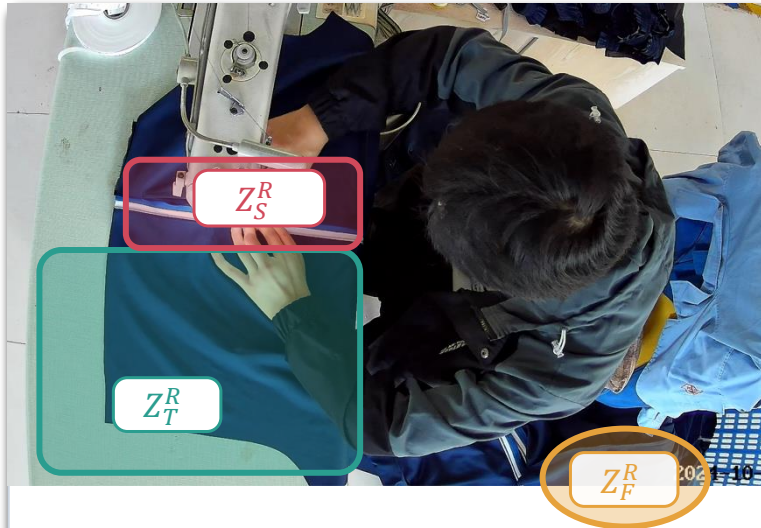


Spatial translation from the real sewing workstation to the VR-WCE

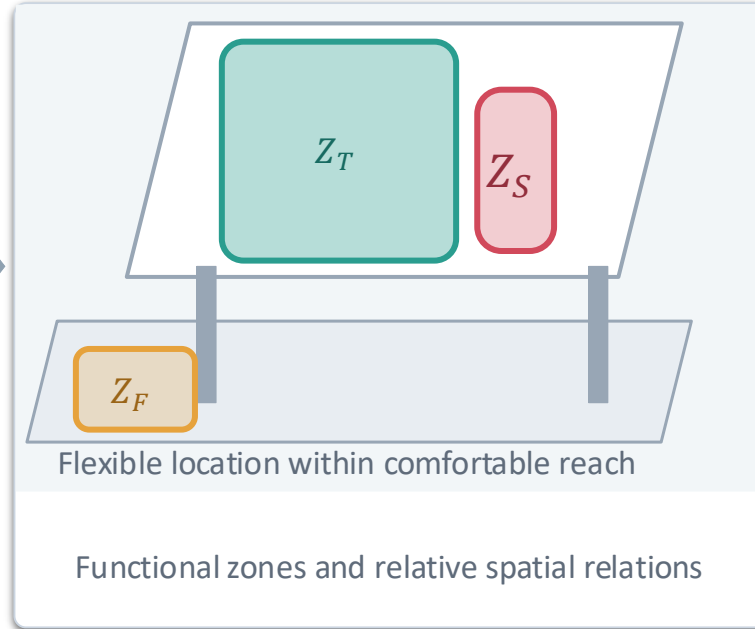
Functional relations were preserved through geometric abstraction rather than exact metric replication.

A Real workplace



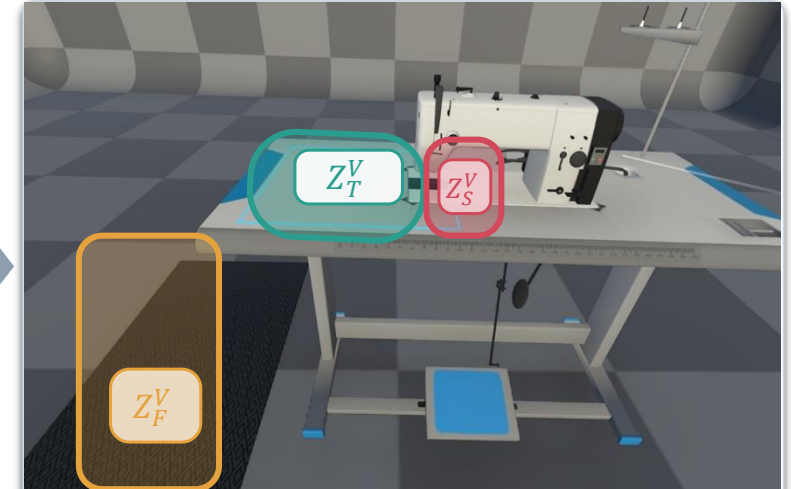
Observed workspace and material locations

B Functional spatial abstraction



Functional zones and relative spatial relations

C VR implementation



Corresponding task regions and spatial references in the VR workstation

Preserved material-flow sequence



Z_F :floor-level material-exchange zone Z_T :tabletop handling zone Z_S :needle-adjacent sewing zone